

PHOTON ENERGY

The photoelectric effect causes electric current to flow from the surface of certain metals when they are bombarded with light. However, it only works when that light is shorter than a certain wavelength; even intense bombardment with longer-wavelength light cannot trigger it. In 1905, Albert Einstein determined that this was because the effect depends on electrons being struck by individual light quanta. Based on his discovery, Einstein argued that all radiation takes the form of quanta or “photons.”

Albert Einstein won his only Nobel Prize for describing the photoelectric effect, not for his theories of relativity.

